

Unsupervised Feature Learning for Classifying Dynamic Tactile Events Using Sparse Coding

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Abstract—Robotic operations that involve the displacement of objects generate different kinds of dynamic events. These may simply correspond to normal robot-related motion, or contact(s) with the object(s) during grasping, but they may also be potentially-problematic events like slippage. In this paper, we use sparse data from tactile sensors to detect slippage and discriminate object-gripper slip from object-world slip. The method we propose can also identify vibrations that correspond to other dynamic events automatically, even when those events are not related to slippage. The tactile data can then be classified, allowing the robot to react accordingly. To achieve this goal, we compute the power spectral density (PSD) of the tactile dynamic signal, and we apply transformations to the PSD that were inspired by the automatic speech recognition (ASR) field. The originality of this work comes from using a sparse representation of the transformed data to obtain sparse vectors containing a small set of high-level features. Those sparse vectors are then used as inputs to a simple linear support vector machine (SVM), that acts as a classifier and quickly estimates the event to which they correspond. Our method was tested on data obtained from 244 experiments that were conducted on 32 different everyday-objects. Results show that we can successfully discriminate most of the dynamic events we studied in this work. Moreover, by using this technique, we are able to detect slippage with an accuracy of 92.60% and to differentiate object-gripper slip from object-world slip with a success rate of 89.42%.

I. INTRODUCTION

In order to better match the tremendous manipulation and grasping skills of humans, robots will need to possess greater intelligence. Examples of such intelligence include the ability to predict grasp quality, and reflexes that allow the robot to react to unwanted slippage of a grasped object during motion. These advanced cognitive capabilities need to be built on a foundation of enhanced sensing capabilities. Artificial vision has already contributed greatly to the improvement of robot grasping and manipulation skills, but since the sense of touch is a crucial aspect of grasping, the full solution entails the use of tactile sensing technology. In recent years, tactile sensors have become one of the most fast-spreading technologies to act as a solution for increasing the level of reliability during handling tasks. This is mainly due to the fact they provide on-the-spot information about the interactions occurring between the robot hand and its environment.

Recent work has shown that handling manipulations can be significantly improved through the use of such sensors. Some examples of tactile sensing contributions are given



Fig. 1. The 32 objects used in the experiments

in [1]–[4]. However, as noted in [5], most of the experiments that are reported in the literature use tactile sensors to get inside-hand *static* pressure information. This is useful for obtaining, among other things, an idea of the grasping configuration of an object inside the gripper, and the force applied to the object [3]. Pressure information is also used in [6] to assess grasp stability and to recognize object instances.

The occurrence of grasped-object slippage is another important event that must be monitored, since it can easily lead to grasp failure during object-handling tasks. Quick and accurate slippage detection could greatly improve handling capabilities, as it could prompt reactive behaviour from the robot. Some researchers have used fast sensing during motion, commonly referred to as *dynamic sensing*, to detect slippage [7], [8]. Slippage detection may seem like a simple task, since the slippage will generate vibrations on the dynamic tactile sensing system that can easily be detected; but in fact, it is a relatively hard problem to solve, as slippage is not the only dynamic event that occurs at the fingertips. Similar dynamic tactile signals are generated by other types of events, making it difficult to determine when slippage is actually occurring. For instance, vibrations may also result from the motion of the robot, or when the robot holds an object and slides it across a surface.

Some researchers recently proposed a method of detecting slippage and distinguishing it from other dynamic tactile events. In [9], [10], the authors used tactile sensors to acquire an array of pressure readings at a high frequency, and built a discriminant marker based on the data's PSD to differentiate

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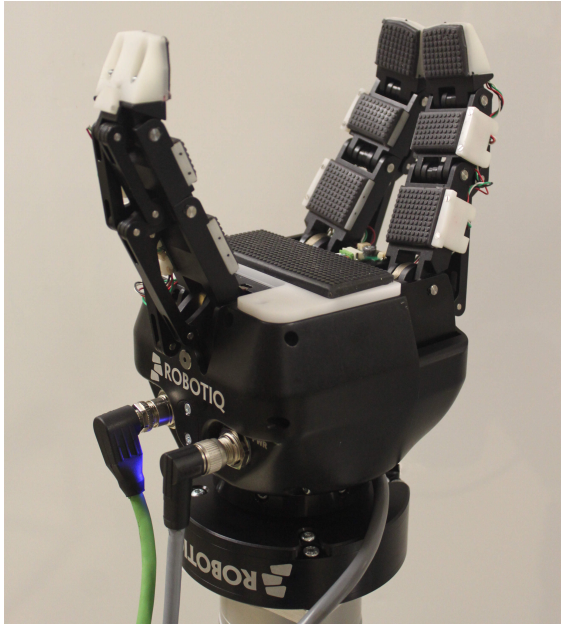


Fig. 2. Tactile sensors integrated to a Robotiq gripper

between two events: the slippage of a grasped object, and an object sliding on an external surface. The authors in [11] used a method based on the analysis of the fast Fourier transform (FFT) of high-frequency pressure data to classify the types of dynamic events into three different groups.

All these researchers used handcrafted features to discriminate between the different classes. The present work instead uses an unsupervised feature-learning approach that gives a high-level representation of the dynamic tactile signals. The signals are then fed to an SVM classifier to distinguish the slippage of an object inside the hands from three other classes of dynamic events. This high-level yet minimal representation is obtained by using sparse coding [12] on the spectrograms of the dynamic data. Examples of sparse coding appear in many fields, including ASR [13], [14], music genre classification [15], image classification [16], handwritten character recognition [17] and text categorization [18]. Nevertheless, to the best of our knowledge, this is the first time sparse coding is applied to tactile data in order to detect slippage and classify dynamic events. The use of an unsupervised feature learning approach, such as sparse coding, eliminates the challenge of arbitrarily choosing the adequate features that will enable the differentiation of our four classes. Sparse coding also allows us to exploit more data, since only the SVM classifier needs labeled data. Indeed, sparse coding itself only requires unlabeled data, which is easy to generate in quantity.

To test our approach, we conducted 244 experiments involving 32 different everyday-objects (shown in Fig. 1). Section II provides more theoretical details about the proposed approach and section III describes the robotic hardware and software components that were used to collect the data. In section IV, we validate our method by performing 10-fold cross validations on the data collected during our experiments, and analyze the impact of sparsity level on

classification accuracy. Finally, in section V, we discuss the results and the future work they might inspire.

II. PROPOSED APPROACH

As mentioned in the introduction, one aim of this work is to distinguish the slippage of a grasped object from three other types of dynamic events. To achieve this goal, we used the dynamic signal of a tactile sensor, which will be presented in the next section. By dynamic signal, we mean a high-frequency signal that is a direct function of the variation of the stress applied at the fingertip. For the purpose of this work, we define a total of four different classes related to dynamic events of interest, which are: Class 1 - An object is slipping relatively to the gripper (Object/gripper slippage); Class 2 - An object in the gripper slips on a surface (Object/world slippage); Class 3 - Null class, i.e. no relevant dynamic events are occurring and Class 4 - External vibrations due to an event not related to slippage. In our case, an example of external vibrations is the one generated by the robot we use (UR5), under certain conditions. Effectively, while experiencing with this apparatus, we found out that using heavy objects ($1.5 - 2 \text{ kg}$) makes the robot arm generating perceptible vibrations, which doesn't occur when handling lighter objects. These vibrations seem to arise from some of the last joint motors when the combined weight of the gripper and the object is close to the robot's payload limit. These vibrations, that are not related to any kind of slippage, thus affect our manipulations when working with some particular objects. This last class could also represent vibrations generated by machines of different kinds in an industrial context.

Fig. 4 depicts four typical pairs of raw data, and the spectrograms related to each one of these classes. Evidently it would be difficult to find a criterion for class discrimination based solely on the raw dynamic data.

From the spectrograms themselves, one may observe clues about the different frequency patterns in each class. For example, object-gripper slippage seems to generate most of its energy in a narrow low-frequency band, whereas object-world slippage produces a wider low-frequency band. In general, the energy seems to be spread slightly more homogeneously across all frequencies when there are no

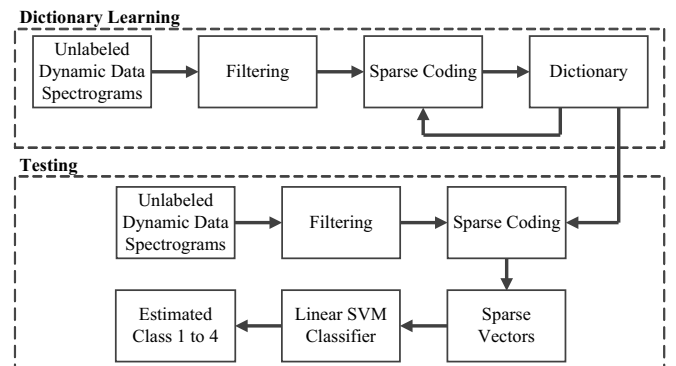


Fig. 3. Functional diagram of our proposed approach

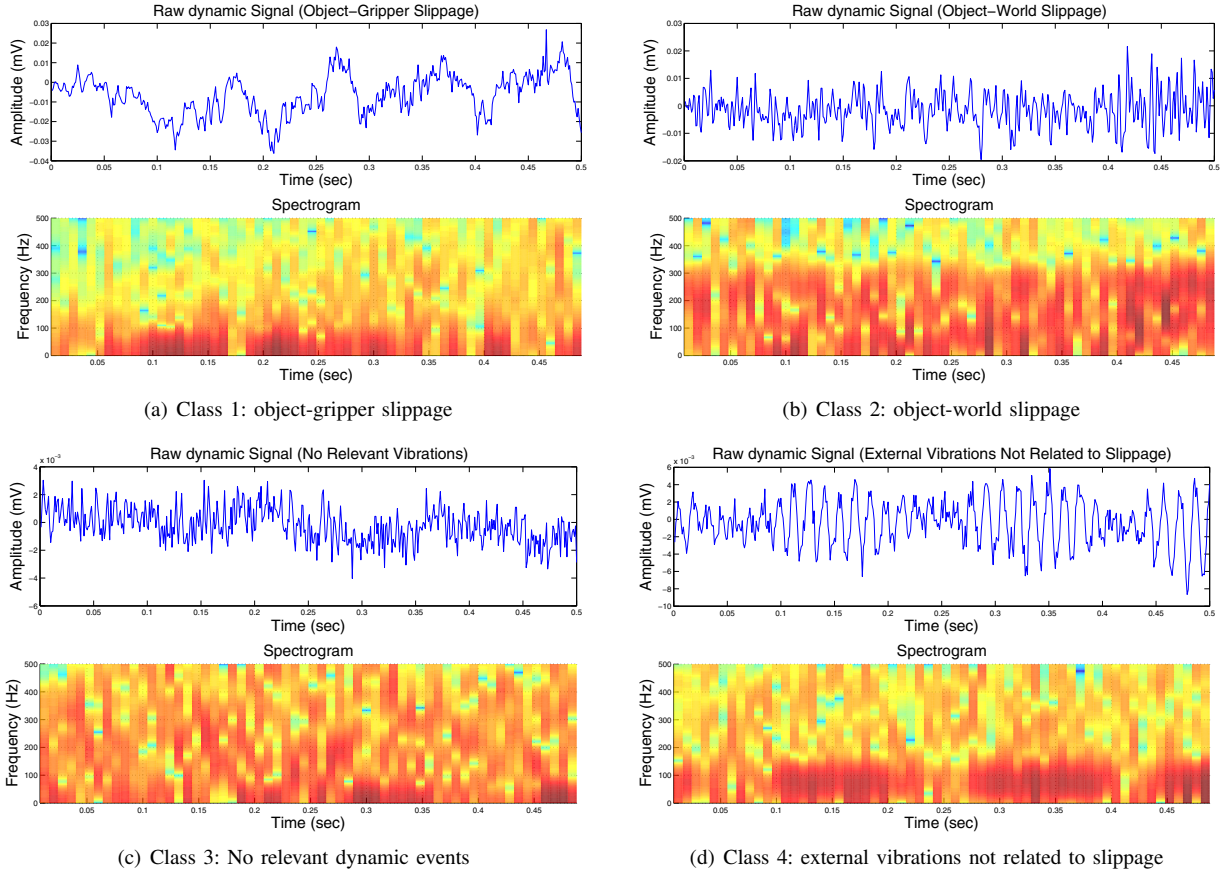


Fig. 4. Raw dynamic data and spectrograms of the four classes

notable dynamic events. Finally, the vibrations that originate from the robot produce a very unique frequency pattern with some periodic darker spots in the low-frequency band. In the spectrogram's space, one can more easily observe patterns for data discrimination.

To produce a sparse representation of the spectrograms that would illuminate these high-level features, we began by building a dictionary of basis using filtered unlabeled dynamic data spectrograms. Once the dictionary was learned, we used it to produce high-level sparse representations of new dynamic spectrograms. The obtained sparse representations are then input to a linear SVM for classification. Fig. 3 shows the basic outline of the proposed method.

A. Pre-processing

As noted by [19], the most important region of the frequency spectrum, while differentiating and detecting textures, is the one that corresponds to frequencies below 250Hz. Humans rely on fast-adapting mechanoreceptors, such as the Meissner and Pacinian corpuscles, to feel textures and slippage. Since these mechanoreceptors have a peak response of, respectively, 30Hz and 250Hz [20], we assume the most relevant frequency modes for discriminating dynamic events are the relatively low-frequency phenomena (below 250Hz). As in many ASR approaches, we applied a log Mel-filter bank (MFB) [21] to reflect the fact that low frequencies are more important to distinguish than high

frequencies. This transformation reproduces the behavior of the human cochlea, which is able to differentiate low frequencies more easily than high frequencies.

B. Sparse Representation of Dynamic Data

In this section, we attempt to find an approximated sparse representation of the obtained log MFB spectrograms that captures high-level features from the data. We use sparse coding to represent the input vectors as a linear combination of the basis vectors that are stored in a dictionary.

Let $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)} \in \mathbb{R}^k$ be the m segments of a certain spectrogram $\mathbf{X} \in \mathbb{R}^{k \times m}$, such that each segment has k values of power over some frequency range. The idea is to find a sparse vector $\boldsymbol{\alpha}^{(i)} \in \mathbb{R}^n$ for each $\mathbf{x}^{(i)}$ by using some *a priori* learned basis from a dictionary $\mathbf{D} := [\mathbf{d}_1, \dots, \mathbf{d}_n] \in \mathbb{R}^{k \times n}$, such that:

$$\mathbf{x}^{(i)} \approx \sum_{j=1}^n \mathbf{d}_j \alpha_j^{(i)} \quad i = 1, \dots, m \quad (1)$$

To obtain the sparse vectors $\boldsymbol{\alpha}^{(1)}, \dots, \boldsymbol{\alpha}^{(m)}$ that capture high-level patterns of $\mathbf{X}$ in the dictionary, the dictionary of basis $\mathbf{D}$ must first be learned. This is done by minimizing the following objective function:

$$\min_{\mathbf{D}, \boldsymbol{\alpha}} \sum_{i=1}^m \left(\left\| \mathbf{x}^{(i)} - \sum_{j=1}^n \mathbf{d}_j \alpha_j^{(i)} \right\|^2 + \lambda \sum_{j=1}^n |\alpha_j^{(i)}| \right) \quad (2)$$

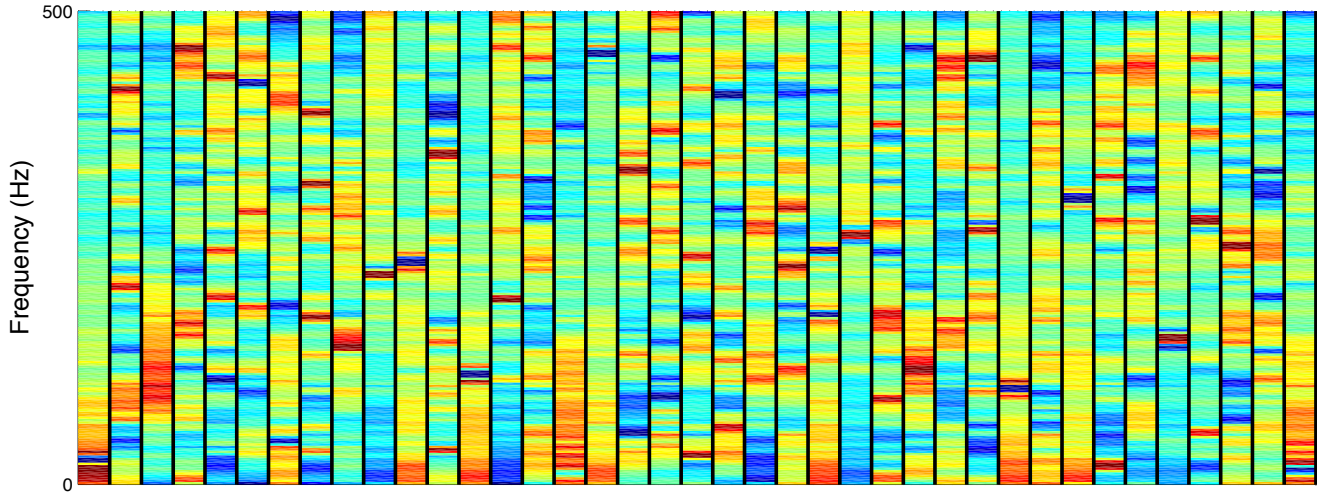


Fig. 5. A sample of the 128-basis dictionary that gave the best results during this work

The first term of equation 2 inside the summation is the squared representation error, thus penalizing the objective function for poor representation of the input vectors. Regarding the second term, λ is an arbitrarily-set scalar that will define the importance of sparsity. This second term penalizes the objective function when non-sparsity is high, and thus is responsible for making each $\alpha^{(i)}$ sparse.

The double minimization problem stated in equation 2, is a complex one that is known to be computationally expensive. However, [22] has shown that it can be split into two convex optimization problems, which can then be solved iteratively. Moreover, dictionary learning can be performed offline so that it does not affect live operations.

Fig. 5 depicts a sample of a 128-basis dictionary that was used during the experiments. Each column corresponds to a basis for spectrogram reproduction, and each row is a frequency bin. To the untrained eye, Fig. 5 may seem to be merely a big patch of noise; but in fact it represents high-level features. For instance, we attempted to use a pure Gaussian-noise generated dictionary instead to reproduce the spectrograms. This did work, to some extent, but we found that the sparsity needed to be very low in order to get a good-quality reproduction. This is due to the fact that a properly-trained dictionary contains specific high-level features and is therefore far from being random noise.

Given such a stable dictionary D , sparse vectors for new input data can be obtained by solving:

$$\min_{\alpha} \left\| \mathbf{x} - \sum_{j=1}^n \mathbf{d}_j \alpha_j \right\|^2 + \lambda \sum_{j=1}^n |\alpha_j| \quad (3)$$

Fig. 6 gives an example of such input-data reconstruction at two different sparsity levels. Even with a considerable sparsity level of 91%, the overall quality of the reproduction is still good, showing that the sparse vectors successfully capture high-level information from the input data. It is this high-level information, rather than the complete spec-

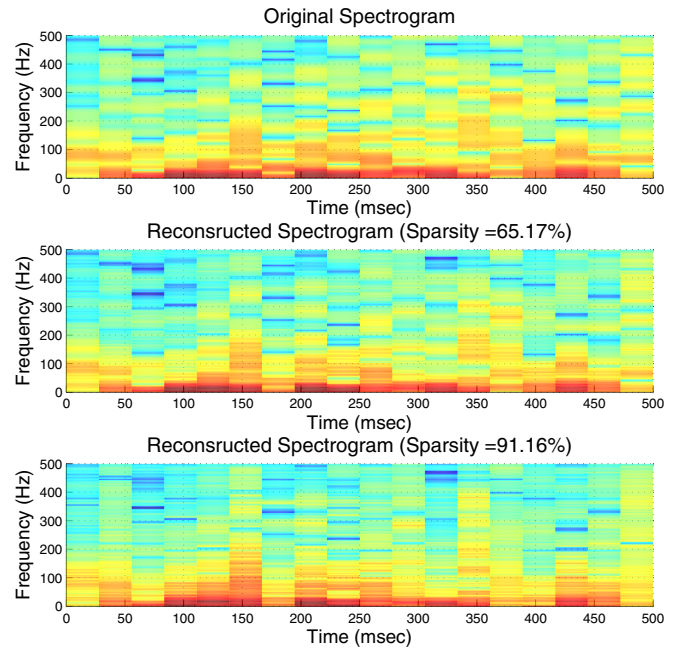


Fig. 6. Reconstruction of a spectrogram at different sparsity levels

trograms, that is used to train and test a simple linear SVM classifier. To do so, we concatenate each sparse vector of a given reconstructed spectrogram and send this information to the SVM.

III. EXPERIMENTAL SETTINGS

A. Hardware description

The tactile sensor technology we used to conduct our experiments is shown in Fig. 2. This sensor was developed in our research laboratory, CoRo, at École de technologie supérieure (ÉTS) and commercialized (patented) by Kinova Inc., and Robotiq Inc. [23]. The sensor can measure both static pressure and dynamic variations of pressure over time. It uses two independent acquisition channels, so it can

acquire both static and dynamic data at the same time and location. The sensor relies on capacitive sensing to acquire pressure data, which is done at a rate of 25 Hz , with a resolution of 3×3 taxels. The sensor's dielectric is composed of two layers of microstructured silicone that are filled with nanoparticles of ferroelectric ceramic. This dielectric allows the sensor to have a wide measurement range as well as a relatively high sensitivity; it reacts to even a minimal pressure of 10^{-4} N per taxel. Dynamic data is acquired with a transimpedance amplifier, rather than by using the difference of capacitance values over time. Any external stimulus causes a change in the sensor's equilibrium conditions; as a result, the sensor is highly sensitive to most dynamic events, such as vibrations or brief contact(s). It has a sampling frequency of 1000 Hz , thus a bandwidth of 500 Hz .

We integrated several of these tactile sensors to a three-finger adaptive robot gripper from Robotiq, Inc.; but in the present work, we rely solely on the dynamic data from a single tactile sensor, that was located on the tip of one of the fingers. The gripper has a maximum payload of 7.5 kg (2.5 kg for each fingertip), and a stroke of 155 mm . In our work, the pinching grasp movement was used for all of the robotic manipulations. As shown in Fig. 2, a force and moment sensor was also attached to the gripper, but it was not used for our experiments.

Our robotic framework consists of a UR5 manipulator from Universal Robot. The UR5 has six independent rotational joints and a maximum payload of 5 kg .

B. Data acquisition

To collect our training and test data, we used the hardware components described in section III-A along with the objects shown in Fig. 1. We performed a total of 244 experiments: 60 experiments with object-gripper slippage (class 1), 64 experiments with object-surface slippage (an object held by the gripper slipped across the surface of a table) (class 2), 60 experiments in which no abnormal vibrations were generated (robot steady or moving a light object) (class 3) and 60 experiments with robot vibrations (a heavy object, held by the robot, caused the robot to vibrate while it was either stationary or mobile) (class 4).

For the first two classes of the latter experiments, all 32 different objects were used at least once. To conduct the object-gripper slippage experiments, the objects were fixed to a clamp with a spring attached. The spring produced an increasing friction force while the robot moved away from the fixation point. At a certain point, this friction force became too high and slippage began to occur. In the case of object-world slippage experiments, we first ensured that the objects were firmly held by the gripper, and then programmed the robot to follow a certain trajectory, thereby causing the object to slip on the table. For the third class, only light objects or no objects were used, while for the fourth class we used heavy objects to generate robot-induced vibrations.

To collect all these data, a ROS framework running under Ubuntu was used to control the gripper and collect the tactile

sensor's dynamic data at 1000 Hz . For each of the 244 experiments, we isolated a relevant 500 ms portion of the data, which allowed us to compute the spectrograms we used to test our method. Those spectrograms were obtained with a Hamming window, sized 50 ms with an overlap of 25 ms . Our sparse coding algorithms were adapted from the MATLAB code available in [22].

IV. EXPERIMENTAL RESULTS

To validate our approach, we first used a 10-fold cross validation on a total of 244 spectrograms data to verify the classification efficiency of all the possible considered events. We used a dictionary that was previously learned using other data than the one considered here. The latter is containing 128 basis and was used to create sparse vectors with an average sparsity of 83%. To perform multiclass classification using binary classifiers like SVMs, we used the well-known one-vs-one strategy, where we count class membership votes across all SVMs. If it happened that the maximum number of votes was shared by more than one classes after classification, then we used the soft margins of the support vectors of each concerned SVMs as a confidence level to determine the winning class. Matrix of confusion shown in Fig. 7 expresses the successes and failures of the classifications for each input sparse vectors. From those results, we clearly see that we are able to detect object-gripper slippage (C1), which corresponds to the critical class in most handling tasks, 86.67% of the time (rounded to 87% in the figure). The matrix also shows that we can perfectly discriminate external vibrations, which in our case were generated from the UR5 robot while in particular conditions. We also note that there is significant confusion between object-world slippage (C2) and the null class (C3). This is most probably due to the fact that in our experiments, many of the 32 objects were not generating any perceptible vibrations from the tactile sensor point-of-view while slipping on the table. For instance, one of the objects was a plush toy, so when it slipped across the table it did not generate any clear vibrations. Furthermore, in Fig. 8, we analyzed discrimination success between each pair of classes, in order to get an indicator of how easy those dynamic events can be discriminated from each others.

Real	Object/hand	0.87	0.10	0.02	0.02
	Object/world	0.09	0.52	0.39	
	Null	0.02	0.42	0.57	
	External vibration				1.00
		Object/hand	Object/world	Null	External vibration
		Classified			

Fig. 7. Confusion matrix of the four different classes

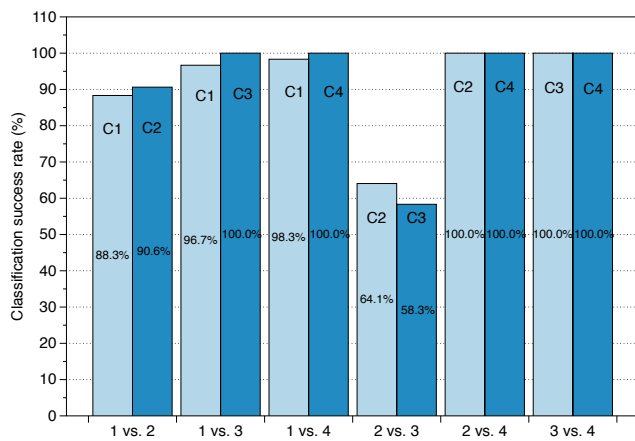


Fig. 8. Histogram showing comparisons between pairs of classes

From this image, we see that most of the discrimination attempts were highly successful (apart from some difficulty with distinguishing between class 2 and class 3). Object-gripper slippage was differentiated from object-world slippage with an overall efficiency of 89.42%. We also observe that almost all of the pairs involving class #4 were perfectly sorted. This is because the vibrations transmitted by the robot during class 4 events are energized at frequencies that are significantly different from the other events we considered. The pairs involving class #1 were also very well sorted. Perhaps this is because class #1 is the only type where the vibrations' origin is located directly on the sensor. By contrast, classes 2 and 3 were differentiated from each other with an overall success rate of just 61.28%, so it might be unreliable to set an automatic discrimination scheme between these two classes. As mentioned before, this relatively low success rate may be due to the low level of vibrations that were generated by some of the objects while they slipped across the table's surface. A significant number of the 32 objects generated vibrations that might not have been strong enough to be perceived clearly by the tactile sensor.

Fig. 9 shows the relation between the sparse vectors' sparsity level and the classification success rate. One can see that if the sparsity is very low, the classification efficiency is

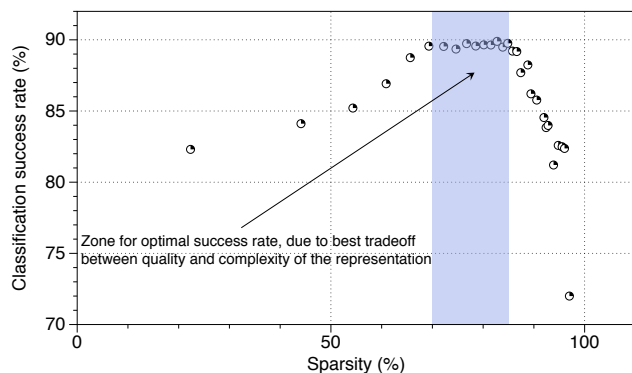


Fig. 9. Classification success rate at different sparsity levels

not optimal due to the fact that the high-level features of the input data are not obvious enough. On the opposite, if the sparsity becomes too high, then we start losing high-level features and classification accuracy also decreases.

To expose the classification success rate of other possible combinations of classes, let us define a set of input sparse vectors $\xi_i \in C_i$ (e.g., $\xi_1 \in C_1$ and $\xi_{1234} \in C_1 \cup C_2 \cup C_3 \cup C_4$), where C_i is the class $i \in \{1, 2, 3, 4\}$ and let D_i be the available classe(s) at the time of classification. Table I shows the resulting success rates when a 10-fold cross validation is performed on each scenario of automatic discrimination. We again have used the one-vs-one strategy, in which class membership votes are counted across all SVMs, to perform multi-class classification. If, after classification, the maximum number of votes happened to be shared by more than one class, we took the soft margins of the support vectors from each of the concerned SVMs as a confidence level to determine the winning class.

TABLE I
CLASSIFICATION VALIDITY ACCORDING TO CONSIDERED CLASSES

	D_{123}	D_{134}	D_{124}	D_{234}	D_{1234}
ξ_1	84.23%	92.17%	83.04%	n/a	86.67%
ξ_2	44.01%	n/a	89.32%	54.47%	51.56%
ξ_3	47.89%	98.93%	n/a	48.29%	56.67%
ξ_4	n/a	100.00%	100.00%	100.00%	100.00%
ξ_{123}	63.07%	n/a	n/a	n/a	n/a
ξ_{134}	n/a	97.78%	n/a	n/a	n/a
ξ_{124}	n/a	n/a	91.96%	n/a	n/a
ξ_{234}	n/a	n/a	n/a	71.78%	n/a
ξ_{1234}	n/a	n/a	n/a	n/a	73.36%

The most general result occurs when the input sparse vectors belong to all four classes (ξ_{1234}) and are also classified into four classes. Considering this scenario, we get an overall classification success rate of 73.36%. In our case, classes 2 and 3 are again the most difficult to distinguish, while class 4 has a perfect score and class 1 has a very good success rate too (86.67%). These results are consistent with those given in Fig. 7, and show that in general, classes 1 and 4 are easily distinguished, whereas classes 2 and 3 are more difficult to differentiate.

However, this situation changes when we stop considering either one or the other of these two classes. For example, if we stop considering class 2 and focus on classes 1, 3, and 4, we get an overall efficiency of 97.78%, and the individual sparse vectors from each of the classes have success rates ranging from 92.17% to 100.00%. The effect is similar when we stop considering class 3 and work only with classes 1, 2, and 4. Finally, as it can be seen from Table I, to efficiently classify the sparse vectors from classes 2 and 3, only one of them may be considered at a time (i.e., when class 2 is being classified, class 3 must be left out of the consideration, and vice versa). Also, one that is interested only in detecting slippage could merge the classes 2, 3 and 4 together and use a single svm that would be used only to discriminate class 1 from the rest. When applying this concept with our data,

we found out that object-gripper slippage was detected with an accuracy of 92.60%.

V. CONCLUSIONS

This work has shown that sparse representations of spectrograms, computed from a tactile sensor's dynamic data, can be used to differentiate and identify certain dynamic events. Moreover, this method can be successfully adopted to detect the slippage of an object while it is held by the gripper. During our tests, this method made possible the distinction between object-gripper slippage and object-world slippage with an accuracy of 89.42%. It also showed that by merging the last three classes, it was possible to detect object-gripper slippage with a success rate of 92.60%. While classes 2 and 3 were problematic when considered together, this challenge may be overcome by either merging these two classes together, or ignoring one of them completely, in order to automatically classify dynamic events with greater accuracy. Future work will concentrate on developing an adaptive dictionary that can exploit the new data to evolve over time.

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